

10 Top AI Model Providers & Releases (2024–2025)

Comparative analysis of leading foundation models, contexts, token costs, and targeted use cases

ASSIGNMENT PART 1

MODEL PROVIDER	MODEL NAME	RELEASE DATE	KNOWLEDGE CUTOFF	CONTEXT WINDOW	COST / 1M TOKENS (IN / OUT)	TYPE	WHEN TO USE THIS MODEL
 OpenAI	GPT-4o	May 2024	Oct 2023	128K	\$2.50 / \$10.00	Proprietary	Complex multimodal tasks, fast conversational response & vision API integrations
 Anthropic	Claude 3.5 Sonnet	Oct 2024	Apr 2024	200K	\$3.00 / \$15.00	Proprietary	Autonomous coding, complex logical reasoning & long-form document synthesis
 Google	Gemini 1.5 Pro	Sep 2024	Nov 2023	2,000K	\$1.25 / \$5.00	Proprietary	Analyzing full repositories, multi-hour audio, raw video & massive context retrieval
 Meta	Llama 3.3 70B	Dec 2024	Dec 2023	128K	Open / Self-Host	Open Weights	Privacy-first self-hosted enterprise deployments & custom domain fine-tuning
 DeepSeek	DeepSeek-R1	Jan 2025	Jul 2024	64K	\$0.55 / \$2.19	Open MoE	High-tier math proof generation, algorithm synthesis & ultra low-cost reasoning
 Mistral AI	Mistral Large 2	Jul 2024	Feb 2024	128K	\$2.00 / \$6.00	Commercial	Multilingual European language workloads & concise high-density code generation
 Cohere	Command R+	Aug 2024	Mar 2024	128K	\$2.50 / \$10.00	Proprietary	Enterprise Retrieval-Augmented Generation (RAG) & multi-step tool call execution
 Microsoft	Phi-3.5 MoE	Aug 2024	Mar 2024	128K	Open / Free	Open SLM	On-device local inference, mobile edge processing & constrained compute budget
 Alibaba	Qwen2.5-72B	Sep 2024	Jun 2024	128K	Open / \$0.40	Open Weights	High-accuracy Asian multilingual processing & cost-efficient open benchmark leader
 xAI	Grok-2	Aug 2024	Apr 2024	128K	\$2.00 / \$10.00	Proprietary	Real-time live web search synthesis & social platform trend sentiment analysis

How Transformers Work: 3 Architectural Pillars

Understanding vector embeddings, self-attention mechanisms, and feed-forward neural networks

ASSIGNMENT PART 2

1. Vector Embedding & Positional Encoding

Tokens are converted into continuous high-dimensional vectors capturing semantic meaning.

- **Token Embeddings:** Maps discrete words/subwords into continuous dense vectors (d_{model} space).
- **Positional Encoding:** Injects sequence order (via sinusoidal functions or RoPE) into inputs, enabling non-recurrent parallel computation.

2. Scaled Dot-Product Self-Attention

Allows tokens to dynamically compute contextual dependency weights relative to every other token in the sequence simultaneously.

- Calculates **Query (Q)**, **Key (K)**, and **Value (V)** matrices via learned linear projections.
- Multi-Head Attention runs multiple attention heads in parallel to capture distinct relational patterns.

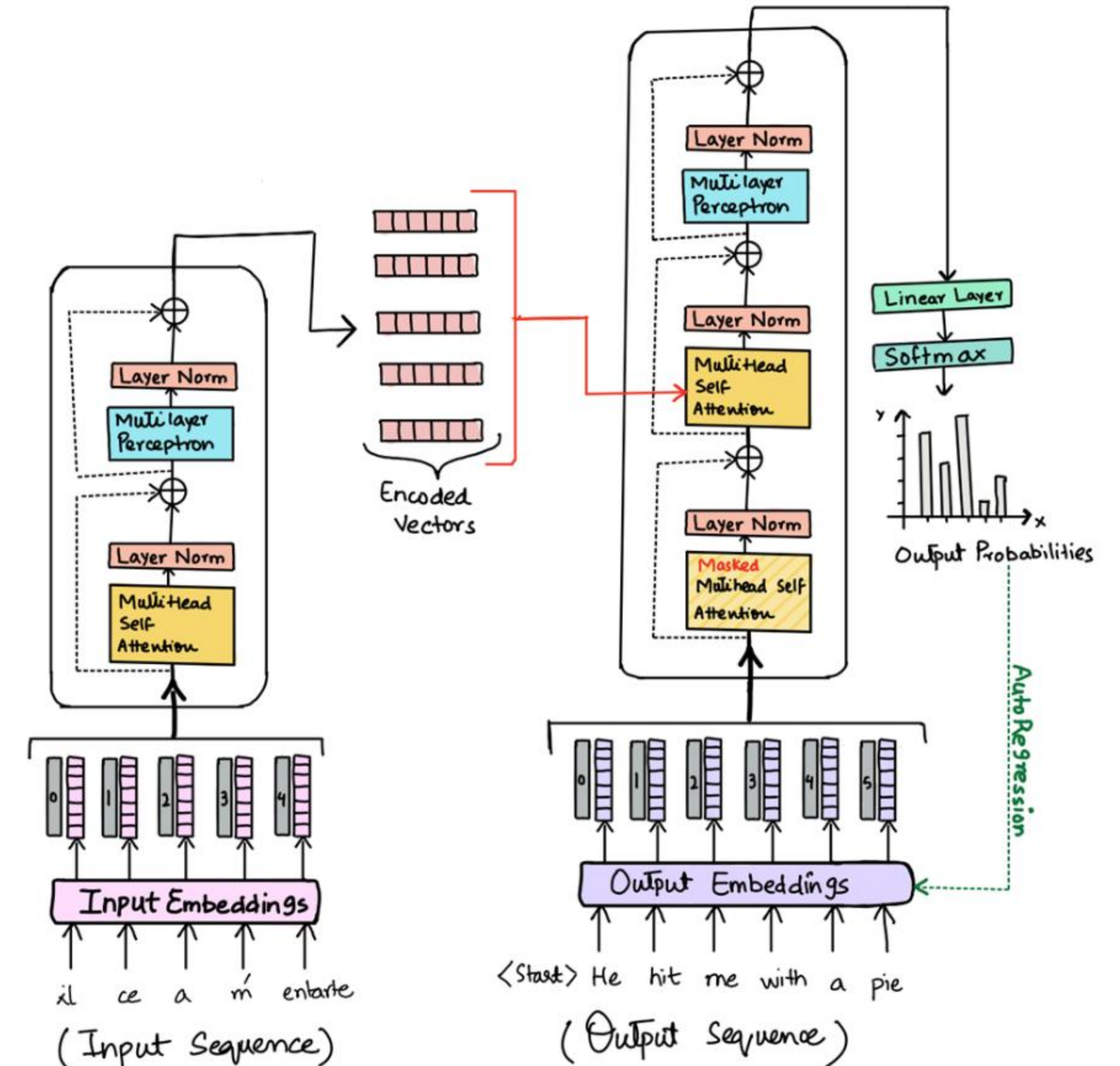
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

3. Position-Wise Feed-Forward Network (FFN)

Applies two dense linear transformations with non-linear activation independently to each position representation.

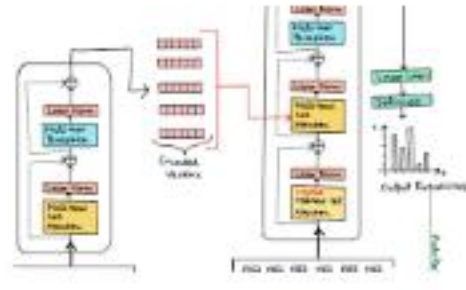
- Processes tokens individually after cross-token attention updates, projecting into higher dimension and back.
- Acts as factual knowledge storage and non-linear feature extractor within the network.

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$



Transformer Block: Encoder-Decoder & Self-Attention Flow Architecture

Image Sources



https://miro.medium.com/1*vrSX_Ku3EmGPyqF_E-2_Vg.png

Source: medium.com